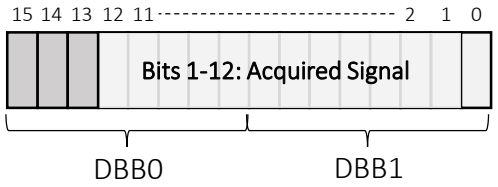
FC5: Acquisition of an Analogue Signal

- In this example, it is supposed that the signal obtained from an analogue input board has been **stored in DBW0 of DB10 (DB10.DBW0)**
- In practice, the signal acquired by the board has been loaded and then transferred in DB10.DBW0
- The board uses a **12-bit A/D and returns a raw data pattern with the following format**



DBB0 is the Most Significant Byte. DBB1 is the Least Significant Byte
Bit 0 (DB10.DBX1.0) reports the status of the acquisition: if DB10.DBX1.0=1, then there has been an acquisition error
Bits 1-12 contain the acquired signal expressed as a two's complement number
Bits 13, 14 and 15 are not used (they have no meaning)

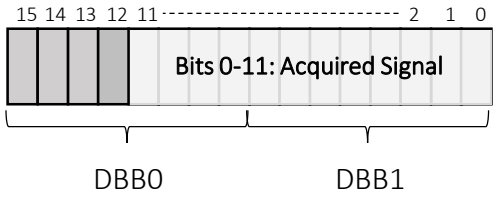
- FC5 has to:
- Check whether or not the **acquisition was successful** (if an acquisition error occurred, then in DB10.DBW6 it must be reported **the HEX constant FFFF**)
- Execute the **conversion in engineering units** and store it in DB10.DBW6
- Execute the **comparison with the threshold values** stored in DB10.DBW2 (high threshold) and DB10.DBW4 (low threshold)
- Prepare a **status byte that reflects the results of the acquisition**
  - The status byte is DB10.DBB8, in which
  - **Bit #0 = 1** if there has been an **acquisition error**
  - **Bit #1 = 1** if the **high threshold has been exceeded**; Bit #2 = 1 if the **low threshold** has been exceeded
  - If the **status byte results ≠ 0**, then in DB10.DBW6 it must be reported **the HEX constant FFFF**

Conversion in Enginering Units

- The expression in engineering units of an analogue signal represents the results of a transduction operation

$$y = f(x)$$

- y is the value of the analog signal expressed in engineering units (temperature, pressure, distance, etc.)
- x is the acquired signal (voltage/current converted in a two's complement sequence of bits)
- f is a dimensional function (from voltage/current to physical variable)
- In the example of FC5, it is supposed that y assumes the value of the acquired signal. In practice, $y = x$ (but y is not a voltage/current).
- x is expressed by the bits 2 – 12. The conversion is then implemented via some shifts that result in the following format



Bits 12-15 must be equal to zero.
Note that this is necessary for positive values. In case of negative values, bits have to be equal to 1 (two's complement representation)
In the example of FC5 it is supposed that values are positive

Input and Output Data

- FC5 does not have any formal parameter. However, there are values/data that are **inputs** to FC5 and other ones that are **outputs**
- DB10.DBW0 is an **input**, since it reflects the value coming from the analogue board
- DB10.DBW2 and DB10.DBW4 are **inputs**. They may be set either by the **user during the initialization phase** or by **other functions that are parts of the PLC program**
- DB10.DBW6 is an output, since it is **calculated and made available by the function**
- DB10.DBB8 is an output for the same reasons as above

New Instructions Introduced by FC5

- **SPB**: **Conditional jump to the specified label**
  - **Depends on RLO** (Jump takes place if RLO=1)
  - **Influences RLO** (if jump does not take place, then RLO is set to 1)
  - **Terminates** the current RLO elaboration
- Note: there exists also an unconditional jump instruction, **SPA** (it does not depend on RLO, does not influence it, does not terminate elaboration)
- **SLW/SRW**: **shift left/shift right of ACCU1 bits,** referring to the **least significant 16 bits (bits 0-15)**. Bits that are shifted out are **replaced by zeros** at the opposite end of ACCU1. **No dependencies on RLO**
- **BEA**: **Unconditional end of a block**. It is the equivalent of a **return** instruction in traditional languages. When such instruction is encountered, the execution of the current block ends immediately and the control of the program flow **is returned to the calling block. No dependencies on RLO**
- Note: there exists also a **conditional end instruction, BEB** for which the **same rules** of the conditional jump instruction apply

FC5 Revisited: FC51

- New version of FC5 with two additional thresholds
- DB10.DBW12 is the HIGH **WARNING** threshold (DB10.DBW2 is the HIGH **FAULT** threshold)
- DB10.DBW14 is the LOW **WARNING** threshold (DB10.DBW4 is the LOW **FAULT** threshold)
- Two additional status bits: DB10.DBX8.3 and DB10.DBX8.4 (overcoming of warning thresholds)
- If warning thresholds are overcome, DB10.DBW6 reports the value of the converted analogue signal
- A further additional bit, DB10.DBX8.7 is reported as cumulative error bit
- DB10.DBX8.7 is 1 if any of the DB10.DBB8 bits is 1

FC51: Value, Thresholds and Status Byte

